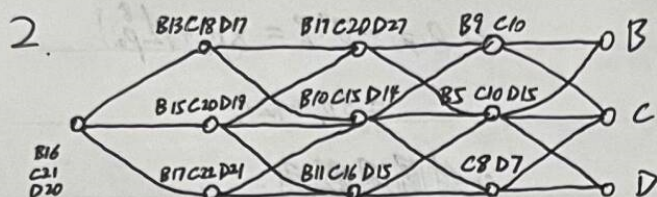


$$\begin{array}{cccc}
 0 & 3 & 6 & 9 \\
 1 & 5 & 4 & 0 \\
 10 & 1 & 0 & 3 \\
 2 & 4 & 6 & 0
 \end{array}
 \rightarrow
 \begin{array}{cccc}
 0 & 2 & 6 & 4 \\
 1 & 4 & 4 & 0 \\
 10 & 0 & 0 & 3 \\
 2 & 3 & 6 & 0
 \end{array}
 \rightarrow
 \begin{array}{cccc}
 0 & 2 & 6 & 10 \\
 0 & 3 & 3 & 0 \\
 10 & 0 & 0 & 4 \\
 1 & 2 & 5 & 0
 \end{array}$$

$$\rightarrow
 \begin{array}{cccc}
 0 & 0 & 4 & 10 \\
 0 & 1 & 1 & 0 \\
 12 & 0 & 0 & 6 \\
 1 & 0 & 3 & 0
 \end{array}$$

$\therefore$ 甲 A Z D 丙 C J B 最少耗时 70
 或 甲 B Z A 丙 C J D



$\therefore$ A 到 B, C, D 最短距离
 分别为 16, 21, 20.

$$\begin{aligned}
 3. \quad f_1(s_1) &= \min_{x_1=s_1} (x_1^p) = s_1^p \\
 f_2(s_2) &= \min_{0 \leq x_2 \leq s_2} (x_2^p) = 2 \cdot \left(\frac{s_2}{2}\right)^p \\
 &\vdots
 \end{aligned}$$

$$f_n(s_n) = n \cdot \left(\frac{s_n}{n}\right)^p \quad \text{最优解 } x_n = \frac{s_n}{n}$$

$$s_{n-1} = s_n - x_n = s_n - \frac{s_n}{n} = \frac{n-1}{n} s_n$$

$$x_{n-1} = \frac{1}{n-1} \cdot s_{n-1} = \frac{1}{n} \cdot s_n$$

$$\text{类推得 } x_1 = \frac{1}{n} s_n$$

$$\therefore s_n = C$$

$$\therefore \min Z = n \cdot \left(\frac{C}{n}\right)^p = \frac{C^p}{n^{p-1}}$$